



BOILEAU GUILLAUME

Data Scientist / AI Engineer | ML, Signal Processing, MLOps-Aware Workflows & Critical Systems

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EXPERIENCE

Software Engineer - AI-Aware Systems, Integration & Production Diagnostics

VEV Platform Services France

📅 2025 – 2026

📍 Valbonne, France

Context: Development and integration of software services for an EV charging CPMS platform connected to operational infrastructure, data flows and hardware systems.

Objective: Support reliable software integration, operational data-flow continuity, system diagnostics and production follow-up for services connected to field equipment.

- **Software development:** Developed and maintained web and backend services using TypeScript, Angular and Node.js in an operational industrial environment.
- **System integration:** Contributed to the integration and maintenance of software services interacting with operational infrastructure and connected equipment.
- **Production diagnostics:** Analysed service behaviour, data exchanges, communication issues and anomalies affecting operational continuity.
- **Data flows:** Implemented, monitored and corrected FTP-based operational data exchange services.
- **Issue analysis:** Investigated incidents involving software behaviour, operational data exchanges and hardware-connected systems.
- **AI systems exposure:** Worked with chatbot and MCP-oriented architectures, with understanding of tool interaction, model integration and application-level AI workflows.
- **Coordination:** Worked with technical stakeholders to clarify issues, follow priorities and support iterative delivery.
- **Tooling:** Used Zenhub, Git-based workflows and collaborative documentation practices for delivery follow-up and technical coordination.

Senior R&D Engineer - Data Science, Signal Processing, ML Validation & Technical Coordination

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

📅 2023 – 2025

📍 Nice, France

Context: Engineering and data-analysis activities for the LISA ESA/NASA space mission, involving complex software chains, signal analysis, simulation workflows, validation campaigns and international coordination.

Objective: Develop, integrate and validate Python-based data science workflows under strong consistency, traceability, performance and verification constraints.

- **Data science workflows:** Developed Python-based workflows for model integration, comparison, validation and statistical analysis in a mission-level scientific context.
- **Software platform:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. [GitLab: CATpyLISA](#)
- **Statistical modelling:** Applied statistical methods, signal-processing techniques and uncertainty analysis to complex low-SNR data environments.
- **Algorithm validation:** Compared model outputs, simulation results and analysis chains to identify deviations, limitations and performance constraints.
- **ML-oriented analysis:** Used Python scientific computing workflows and ML-aware methods to structure model comparison, anomaly detection and performance interpretation.
- **Critical systems context:** Worked in a high-rigor space mission environment involving traceability, technical evidence, review logic and mission-level performance objectives.
- **Technical studies:** Analysed sensitivity to modelling assumptions, numerical parameters, data-processing choices and configuration changes.
- **Validation logic:** Structured comparison campaigns, consistency checks, acceptance logic and validation evidence for technical outputs.
- **Technical reporting:** Produced analysis notes, validation summaries, technical documentation and review material for contributors and project stakeholders.
- **Coordination:** Coordinated contributors, supervised interns and supported technical activities across multiple stakeholders in an international environment.

Data Scientist - Statistical Modelling, Noise Diagnostics & Performance Analysis

Universiteit Antwerpen, Belgium

📅 2021 – 2023

📍 Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments in a high-requirement international environment linked to gravitational-wave detector performance.

Objective: Identify, characterize and mitigate factors affecting system sensitivity using statistical modelling, signal processing and validation-oriented methods.

- **Statistical modelling:** Developed Python-based pipelines for noise source identification, uncertainty estimation, Bayesian inference and statistical diagnostics.
- **Signal processing:** Analysed instrumental and environmental effects affecting measurement quality, sensitivity and stability.
- **Anomaly detection:** Investigated abnormal behaviours, dominant contributors, weak points and possible mitigation directions.
- **Performance analysis:** Identified contributors to performance degradation and supported prioritisation of correction or mitigation actions.

PROFILE FOR THALES COR-TAIX

Data Scientist / AI Engineer with strong experience in statistical modelling, signal processing, Python-based data science, simulation workflows and algorithm validation for complex and critical systems. Background developed through major space and gravitational-wave collaborations, complemented by recent industrial software experience involving operational data flows, software integration and production diagnostics.

Experienced in **Python, statistics, Machine Learning, TensorFlow, PyTorch, Java, GitLab, Docker**, scientific computing, signal processing, anomaly analysis, model comparison, validation workflows and technical reporting. Also familiar with **chatbot architectures, MCP-based tool interaction**, AI agents, tool-calling logic and model integration into application workflows.

For a **Data Scientist / Operational Research** role in a critical Defence and Aerospace AI environment, I bring a strong analytical profile: analysing complex data, selecting relevant models, developing and validating algorithms, communicating results clearly and contributing to AI solutions under traceability, security and operational constraints.

FIT FOR DATA SCIENCE / AI DELIVERY

- Strong background in statistics, Bayesian inference, signal processing, uncertainty analysis and complex data interpretation.
- Hands-on experience with Python-based data science workflows, scientific computing, model comparison and validation logic.
- Experience with TensorFlow, PyTorch, Java, GitLab, Docker and reproducible software environments.
- Ability to develop, test and validate algorithms using statistical, mathematical and Machine Learning methods.
- Experience analysing weak signals, noisy datasets, anomalies, degradation patterns and performance limitations.
- Familiarity with AI system architectures, chatbot systems, MCP, tool-calling logic and model integration into application workflows.
- Comfortable communicating complex technical results to mixed technical and non-technical stakeholders.
- Strong fit for defence, aerospace and critical-system contexts requiring rigor, autonomy, technical evidence and clear reporting.

TEAM MANAGEMENT

- Supervision of **3 Master's thesis interns (M2)** during software, modelling and data-analysis activities.
- Coordination of technical activities involving **research engineers and technicians** on a **data processing and validation pipeline** for the **LISA** space mission.
- Experience organizing tasks, supporting contributors, reviewing outputs and maintaining alignment across collaborative technical developments.

LANGUAGES

English ● ● ● ● ●

Spanish ● ● ● ● ●

EDUCATION

- **Data-driven decisions:** Analysed measurement and simulation outputs to extract meaningful conclusions from complex technical datasets.
- **Reproducibility:** Produced traceable, reproducible and documented analyses for international scientific and engineering stakeholders.
- **Communication:** Communicated complex results through structured reporting, presentations and collaborative review.

Junior R&D Engineer - Signal Analysis, Simulation & Algorithmic Modelling

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021

Nice, France

Context: Data-analysis and simulation activities for gravitational-wave mission preparation, involving weak-signal detection, modelling, simulation and noise characterization.

Objective: Develop robust analysis methods for complex signals and support technical investigations in demanding system environments.

- **Signal analysis:** Developed methods for the separation of overlapping low-SNR signals in complex datasets.
- **Simulation workflows:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- **Algorithm assessment:** Compared simulated outputs, model behaviours and analysis results to assess consistency, robustness and performance limitations.
- **Technical investigations:** Investigated noise sources through cross-sensor correlation analyses in diagnostic contexts.
- **Scientific computing:** Developed strong expertise in Python-based analysis, modelling, documentation and reproducible workflows.

PROFESSIONAL TRAINING

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI - Andrew Ng

2020

Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

2025

France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, Java, algorithms and software engineering fundamentals.

PROJECTS

CNES / LISA – Data Science, Signal Processing & Algorithm Validation

ESA/NASA Space Mission - Large-scale International Collaboration

2023 – 2025

Nice, France

Contribution to Python-based data science, signal-processing, model-comparison and validation workflows for the LISA space mission within a large international engineering and scientific collaboration.

Data science relevance: Statistical modelling, uncertainty analysis, weak-signal analysis, simulation workflows, model comparison, anomaly investigation and validation evidence.

AI delivery relevance: Development of reusable Python workflows, GitLab-based software platform, reproducible environments, technical documentation and structured review material.

Critical systems relevance: Mission-level performance analysis, traceability, technical evidence, rigorous validation and work in a high-constraint international aerospace context.

Einstein Telescope / LIGO–Virgo–KAGRA – Statistical Diagnostics & Detector Performance

International Collaborations

2021 – 2023

Antwerp, Belgium

Contribution to collaborative developments in data analysis, signal characterization, statistical diagnostics and performance investigations within large-scale international scientific and engineering collaborations.

Role: Development of analysis tools, contribution to technical studies, reporting and collaborative review in high-standard environments.

System impact: Studies on environmental and instrumental noise sources supporting the identification of performance limitations and design constraints for next-generation gravitational-wave observatories.

STRENGTHS

Data science, ML & AI

- Data Science
- Machine Learning
- Deep Learning
- Statistical Modelling
- Bayesian Inference
- MCMC
- Regression
- Classification
- Feature Engineering
- Model Selection
- Model Validation
- Algorithm Design
- Operational Research
- Optimization
- Decision Support
- Explainable AI Awareness
- Ethical AI Awareness

Signal, detection & complex data

PhD in Physics

Université Côte d'Azur

2018 – 2021

Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

2016 – 2018

Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

2017 – 2018

Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016

Orsay, France

Signal Processing Weak Signal Detection Low-SNR Analysis Noise Modelling Anomaly Detection Pattern Recognition Data Fusion Awareness Simulation Workflows
Synthetic Data Uncertainty Analysis Sensitivity Analysis Performance Assessment Complex Data Analysis Scientific Computing

AI engineering & integration

AI System Architecture Model Integration AI Agents MCP Chatbot Architecture LLM Integration Tool-Calling Prompt Engineering Application Workflows
API Integration Data Pipelines Reproducible Workflows MLOps Awareness Deployment Awareness User-Facing AI Systems

Programming, frameworks & tools

Python TensorFlow PyTorch Java Pandas NumPy Scikit-Learn SQL Bash Linux Git GitLab GitHub Docker CI/CD Jupyter
REST API OpenSearch Confluence JIRA

Critical systems & defence context

Critical Systems Defence AI Interest Aerospace Context Space Systems Mission-Level Performance Embedded AI Awareness Frugal AI Awareness
Cybersecurity Awareness Sensitive Data Awareness System Validation Technical Evidence Traceability Risk Identification Robustness Reliability

Collaboration, delivery & communication

Technical Workshops Client-Facing Communication Scientific Communication Technical Reporting Knowledge Sharing Stakeholder Alignment Cross-Functional Coordination
Agile Environment Technical English Autonomy Initiative Problem Solving Analytical Thinking Pedagogy Innovation Mindset